

Data Sources and Methods

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Weeks 4 Progress Update:

(9/8/2026 - 9/14/2026)

Focus

Reviewed literature for environmental factors to include in the species distribution model.

Candidate Variables

The following variables are currently selected based off representation in literature, perceived data availability, or an effort to maintain cross-comparison compatibility with the HSI derived in Geselbracht et al. (2024).

Table 1: Preliminary Environmental Variables Considered for the Eastern Oyster SDM

Variable	Reason	Reference	Data Source
Salinity	Widely available, large driver of ecosystem suitability	(Peyre et al. 2021; Geselbracht et al. 2024; Pollack et al. 2012; Barneset al. 2007)	Water Quality Portal (DeCicco et al. 2026)
Water temperature	Widely available, large driver of ecosystem suitability	(Peyre et al. 2021; Geselbracht et al. 2024; Pollack et al. 2012; Barneset al. 2007)	Water Quality Portal (DeCicco et al. 2026)
Substrate / bottom type	suitable substrate is required for recruitment	(Peyre et al. 2021; Geselbracht et al. 2024; Barnes et al. 2007)	TBD

Table 1: Preliminary Environmental Variables Considered for the Eastern Oyster SDM

Variable	Reason	Reference	Data Source
Water depth / bathymetry	Depth is a key driver in oyster habitat extent	(Peyre et al. 2021; Geselbracht et al. 2024; Pollack et al. 2012; Barnes et al. 2007)	National Centers for Environmental Information (National Centers for Environmental Information 2018)
Dissolved oxygen	Low DO can cause oyster stress and mortality	(Peyre et al. 2021; Geselbracht et al. 2024; Pollack et al. 2012)	Water Quality Portal (DeCicco et al. 2026)
Chlorophyll-a	Common proxy for food resources supporting oyster growth	(Peyre et al. 2021; Geselbracht et al. 2024)	Water Quality Portal (DeCicco et al. 2026)
Oyster bed locations	important for SDM modeling, which requires species presence data	(Peyre et al. 2021; Geselbracht et al. 2024)	Florida Fish and Wildlife Conservation Commission (Florida Fish and Wildlife Conservation Commission 2015)

Week 5 Progress Update:

(9/15/2026 - 9/21/2026)

Focus

Review literature relating to the underlying logic used in MaxEnt species distribution model.

Maximum entropy modeling of species geographic distributions

Authors: Phillips et al. (2006)

Background

The authors introduce Maxent as a method for species distribution modeling when only presence data are available. The authors note that for many species having both presence and absence data is very unlikely. Maxent uses environmental characteristics of known locations and is able to apply them to unknown locations to predict probability of species occurrence.

Niche-based models can predict the geographic extent but will often overestimate the reality of the species distribution. It is recognized that because an environment might fall within the

niche of a given species, doesn't necessarily mean the species are actually established there due to a multitude of geographic, biotic, or anthropogenic barriers. Additionally, there are many assumptions made in such modeling techniques, so the author's express that much care must be taken when choosing which environmental factors to use. With so many assumptions, the authors cite many sources of errors that can be introduced if care is not taken such as using improper temporal or spatial resolutions and scales. Further, the authors state significant errors can also arise from improper data manipulation, poorly optimized environmental models, and interpolation of lower-resolution data.

Finally, the authors acknowledge that there are problems with presence-only data as a whole that must be considered when designing the model. The author's mention things such as geographic sampling bias, spatial autocorrelation, poor sampling design (e.g. small sample sizes and erroneous sample collection/reporting), and missing or important environmental predictors to name a few.

Maxent

The authors describe the basics of maximum entropy modeling where the goal is to estimate an unknown probability distribution while making as few assumptions as possible. As hinted in the name, the model seeks the distribution that has the maximum entropy (i.e. the most uniform distribution), while maintaining the influence of what is known from the presence data. In species modeling, the model is made up of 3 major parts: 1) the study area pixel grid represents all possible location, 2) known presence observations provide sample locations, and 3) the environmental predictor variables act as features that describe the conditions at the species occurrence locations.

Maxent Advantages:

- Only requires presence data;
- Can use both continuous and categorical data, and incorporate interactions between variables;
- Efficient deterministic algorithms have been developed which are guaranteed to converge on optimal probability distribution;
- The Maxent probability distribution has a well described mathematical definition;
- Over-fitting can be avoided by using regularization techniques;
- Provides flexibility for adding thresholds to continuous predictions;
- Can work on relatively limited data and be applied to all levels of scale.

Maxent Disadvantages:

- At the time of the paper, the Maxent method was still new;
- model guidance was still under development;
- uncertainty estimation was not as well developed when compared to other statistical methods;

- overestimation of environmental variables is possible due to the use of an exponential model for probabilities;
- specialized software requirements

In the paper’s methods section, the author’s detail the maxent framework. Principally, maxent relies on the assumption that without outside influence, a distribution would become completely uniform, and in the context of species distribution result in a species having the same probability of occurring at any given geographic location. However, species typically occupy an ecological niche which is defined by a series of environmental constraints. As such, Maxent focuses on satisfying each of the constraints (as bounded using known conditions at known locations), while also maximizing the entropy. To do this, the model uses environmental variables as features and tries to use these features to predict a distribution that resembles that of the observed values at occurrence locations. To prevent overfitting, maxent uses regularization which allows some flexibility between the predicted and observed environmental variables.

Maxent has the ability to transform predictor variables in a number of ways: linear features represent the variable directly (i.e. $f(x) = \text{Temperature}$), quadratic features use a squared predictor allowing for an exponential change (e.g. $f(x) = \text{Temperature}^2$), product features allow for variable interactions (e.g. $f(x) = \text{Temperature} \times \text{Salinity}$), threshold features allows for predictor values to require certain conditions be met (e.g. $f(x) = \text{Salinity} > 20$), and binary features which allows predictor variables to represent categorical factors. The ability to transform variables in such away means that maxent has the capability of handling both continuous and categorical variables.

Keyfindings:

In the rest of the paper, the authors describe an implementation of the maxent model and compare it to another more commonly used presence-only modeling method known as a Genetic Algorithm for Rule-Set Prediction (GARP). Using the same predictor data sets and 70/30 training split, both models were able to perform better than random. A threshold-dependent and threshold-independent evaluation was conducted on both models. Threshold-dependent considers the percentage of known points misclassified as unsuitable (omission rate) and the percentage of overall study region that was classified as suitable (predicted area). Threshold-independent receiver operating characteristic (ROC) and area under the ROC curve (AUC) which can describe the overall model performance (e.g. $AUC = 0.5$ roughly equals to random).

The results of the threshold-dependent evaluation showed that both models produced low omission rates. In the threshold-independent test both models AUC values were significantly greater than random, and Maxent generally had higher AUC values than GARP. Overall, maxent proved to be a stronger presence-only modeling approach.

Relevance to capstone:

Phillips et al. (2006) developed the foundational framework of the proposed Maxent species distribution model. They successfully demonstrate using presence-only species data in combination with spatial environmental datasets to predict habitat suitability. They also outline many

limitations that should be considered when building the model, selecting predictor variables, and when interpreting model outputs. Finally, the authors provide a useful case study that can help guide my own implementation of the model, describing many useful data manipulation techniques as well as model output evaluation and interpretation guidance.

Next Steps

Review literature on proper spatial modeling techniques to convert environmental point data into continuous surface rasters using interpolation techniques such as kriging, etc.

References

- Barnes, T. K., A. K. Volety, K. Chartier, F. J. Mazzotti, and L. Pearlstine. 2007. "A Habitat Suitability Index Model for the Eastern Oyster (*Crassostrea virginica*), a Tool for Restoration of the Caloosahatchee Estuary, Florida." *Journal of Shellfish Research* 26 (4): 949–59. [https://doi.org/10.2983/0730-8000\(2007\)26%5B949:AHSIMF%5D2.0.CO;2](https://doi.org/10.2983/0730-8000(2007)26%5B949:AHSIMF%5D2.0.CO;2).
- DeCicco, Laura, Robert Hirsch, David Lorenz, et al. 2026. *dataRetrieval: R Packages for Discovering and Retrieving Water Data Available from u.s. Federal Hydrologic Web Services*. Version 2.7.25. U.S. Geological Survey; U.S. Geological Survey. <https://doi.org/10.5066/P9X4L3GE>.
- Florida Fish and Wildlife Conservation Commission. 2015. "Oyster Beds in Florida." <https://geodata.myfwc.com/datasets/myfwc::oyster-beds-in-florida/about>.
- Geselbracht, L., M. Johnston, B. M. DeAngelis, and A. Birch. 2024. "Estuary-Specific and Adaptive Habitat Suitability Index Model for the Eastern Oyster *Crassostrea Virginica* in the Pensacola Bay System, Florida, USA." *Coastal Management* 52: 17–34. <https://doi.org/10.1080/08920753.2024.2335129>.
- National Centers for Environmental Information. 2018. *NOAA NOS Estuarine Bathymetry - Pensacola Bay (G130)*. <https://doi.org/10.7289/V5XK8CWK>.
- Peyre, Megan K. La, Danielle A. Marshall, and Shaye E. Sable. 2021. "Oyster Model Inventory: Identifying Critical Data and Modeling Approaches to Support Restoration of Oyster Reefs in Coastal u.s. Gulf of Mexico Waters." In *Diversity and Distributions*, vol. 17. U.S. Geological Survey. <https://doi.org/10.3133/ofr20211063>.
- Phillips, Steven J., Robert P. Anderson, and Robert E. Schapire. 2006. "Maximum Entropy Modeling of Species Geographic Distributions." *Ecological Modelling* 190 (January): 231–59. <https://doi.org/10.1016/j.ecolmodel.2005.03.026>.

Pollack, Jennifer Beseres, Andrew Cleveland, Terence A. Palmer, Anthony S. Reisinger, and Paul A. Montagna. 2012. “A Restoration Suitability Index Model for the Eastern Oyster (*Crassostrea Virginica*) in the Mission-Aransas Estuary, TX, USA.” *PLoS ONE* 7 (July). <https://doi.org/10.1371/journal.pone.0040839>.